

Nicole Swierstra

Computer Science Student, experienced in design, firmware, and software development of digital embedded circuits and systems.

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Website Link

Relevant Skills

Embedded/Firmware programming: ARM/RISCV assembly, c (CMSIS, ChibiOS, FreeRTOS)

Application-side programming: c (Win32), c++ (QT, open-GL, cuda), c#, java, python, Embedded Linux

Design & Simulation: Altium, KiCAD, Fusion360, python, CATIA, SolidWorks, Excel, MATLAB, Simulink

Education

Western Washington University (2022 -)

Computer Science Major, Music minor

Projects & Applied Experience

Electronics Member and Head Firmware Developer, Formula SAE (fall 2023 -):

- **V67 VCU OS** – Created an overhauled version of the V66 VCU operating system, while training new members with some smaller driver projects. With this new VCU codebase, both the critical safety checks and driver control were able to be performed in 5us. Created a series of over 80 tests for verification of the VCU's behavior before every merge to avoid breaking changes. This operating system was a small but meaningful part of designing a robust car that was able to pass technical inspection quickly and ultimately get 17th in FSAE EV 2026.
- **V67 Data & Simulations** – Developed tools for quickly parsing data gotten from test days, including a program that could quickly parse and show data and events where and when they happened. Used this tool to diagnose several issues with the car, including a faulty current sensor, an inverter communication issue and a short in our inverter. Developed and verified a simulation to predict our car's time in the acceleration event and worked on a vehicle dynamics model to predict our time and points in other events.
- **V67 Dash** – Created a very simple dashboard for a driver interface and used it as a teaching opportunity for new members on how to program embedded devices.
- **V66 BMS Revitalization** – Rerouted and reprogrammed a past car's custom BMS to get it running again after errors in the design, manufacturing, and programming lead to an unreliable system.
- **V66 VCU OS** – Developed a low latency control system for an electric car under heavy memory and processing restrictions with a simple custom real time OS. Critical safety checks and driver control were both able to be performed every 5ms, and the operating system code itself had a very small memory footprint of around 940 bytes.

More information on projects can be found on the portfolio page of my website

Employment

Development Intern, Autosport Labs (spring 2024 – winter 2026)

Contributed to various small projects as a part-time contractor, including python modules used application side and embedded projects with chibi-os on the embedded side.